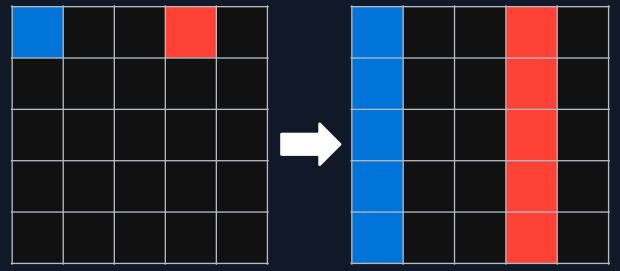
Machine reasoning and ARC-AGI

Can an AI agent learn your world on day one?

Govind Tatachari, CognitivSys LLC * November 2026



A few cells in, a rule inferred, a new grid out.

Use case: an AI agent on day one at a new job

The scenario

A bank hires an AI agent for its financial anomaly-detection team. On day one it meets:

- ledgers and tools it has never seen, and no manual
- a new fraud pattern no rule describes
- a handful of alerts that analysts already labelled

THE REAL QUESTION

Can it infer the new rule from a few examples and apply it correctly, cheaply, today?

Why traditional anomaly detection struggles

- **Static rules and thresholds** are predictable and go stale, so criminals adapt around them [9].
- **Noise:** industry sources put false positives at 85–95% of alerts [9].
- **Drift:** fraud patterns evolve faster than models and rules are retrained [7, 8].
- **Few labels, rare events:** severe class imbalance and limited labelled data [8].

This is exactly what ARC-AGI measures: learn a new rule from a few examples, then act on a new case.

Sources: [7] Hilal et al. 2022; [8] Yang et al. 2026; [9] industry transaction-monitoring analyses.

Idea behind ARC: skill-acquisition efficiency

Can machines imitate the fluid intelligence of Humans?

Chollet (2019): intelligence is how efficiently a system turns a little experience into competence on new tasks, not how much it already knows [1].

ARC-AGI tasks use only Core Knowledge priors: objects, counting, symmetry, goals. No language, no trivia.

ARC-AGI	Day-one agent	Anomaly detection
Demonstration pairs	Examples a colleague shows	Analyst-labelled alerts
A novel task	A new system or rule	A new fraud typology
Exact-match answer	The correct decision	Correct flag or clear
Two attempts allowed	One retry on review	Escalate to analyst
Cost per task	Tokens, time	Analyst-hours saved

Honest scope: ARC-AGI measures the capability (adapting to the unfamiliar). It does not measure financial aspects. A later slide shows how to test both.

Sources: [1] Chollet, On the Measure of Intelligence (2019).

What this talk covers (Intro to ARC-AGI)

1

Try it yourself

A 60-second puzzle: infer the rule from three examples.

2

Why it matters

ARC-AGI measures skill-acquisition efficiency. Two graphs show how current frontier models fared on ARC-AGI-1 and ARC-AGI-2.

3

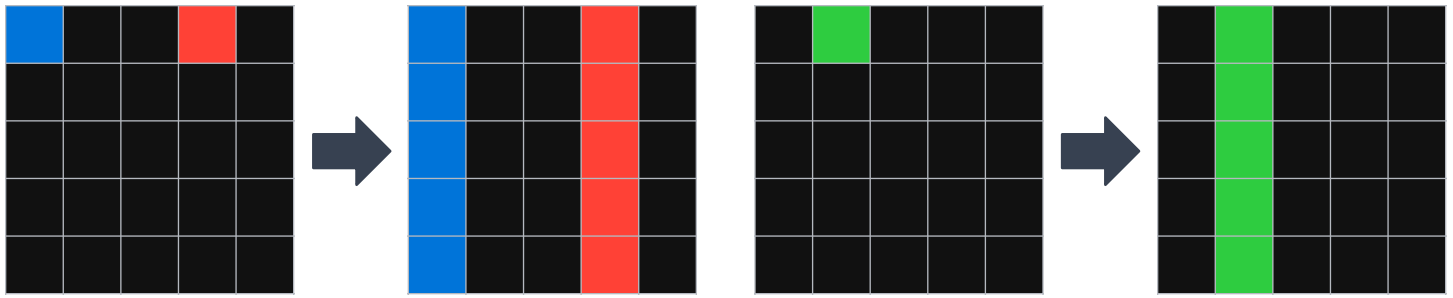
How to apply it

Turn your use case into an ARC-style benchmark and run it on an LLM.

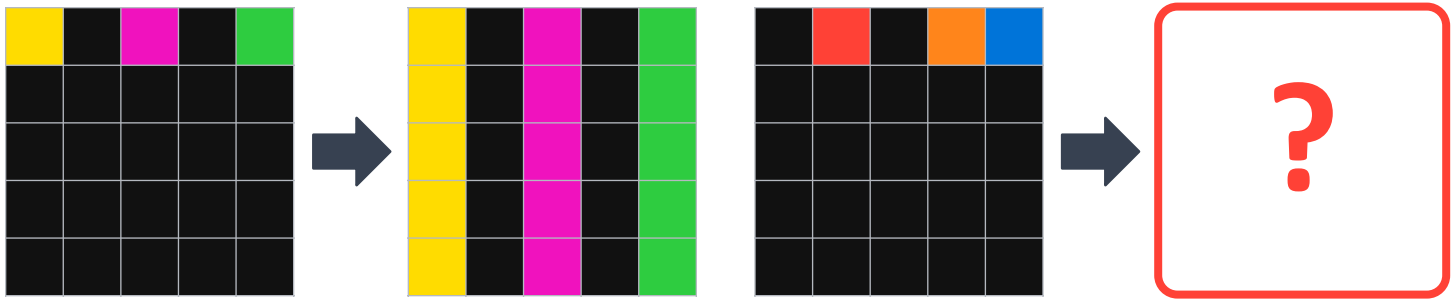
We will use "the day-one anomaly-detection agent" use case throughout this talk.

Try it: infer the rule

Examples



Your turn: the test input



Demo 1: infer-the-rule.html

Open the interactive page (no network needed). Paint the answer, check it, then see what the model produced for the same puzzle.

Notice what you just did

You inferred a rule from three examples and applied it to something new. No retraining.

Methodology: apply ARC-AGI to the use case



ARC-AGI-1 task JSON

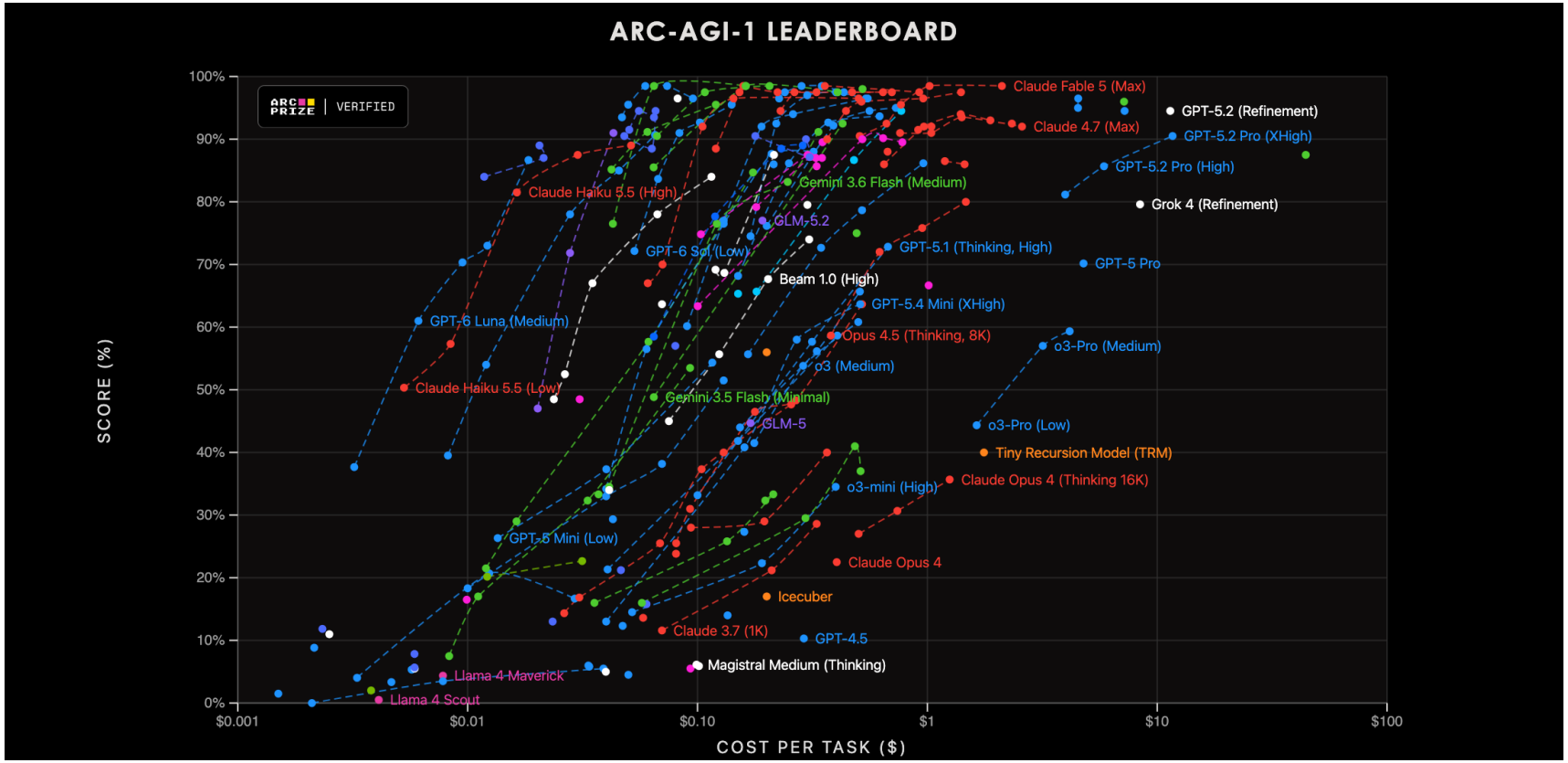
ARC-AGI-1 benchmark tests are encoded in JSON format.

3dc255db.JSON


```
{"train": [
{"input": [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 4, 4, 0, 0, 0, 0, 0, 0, 0, 0], [6, 0, 0, 6, 4, 4, 0, 0, 0, 0,
0, 0], [0, 0, 6, 0, 4, 4, 4, 0, 0, 0, 0, 0], [0, 0, 4, 4, 4, 4, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 3, 3, 3, 3, 0, 0], [0, 0, 0, 0, 3, 3, 7, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 3, 3, 0, 7, 0, 0, 0, 0], [0, 0, 0, 3, 3, 3, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 3, 3, 3, 0, 0, 0]],
"output": [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 4, 4, 0, 0, 0, 0, 0, 0, 0, 0], [6, 0, 0, 0, 4, 4, 0, 0, 0, 0,
0, 0], [0, 0, 0, 0, 4, 4, 4, 6, 6, 0, 0, 0], [0, 0, 4, 4, 4, 4, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 3, 3, 3, 3, 0, 0], [0, 0, 0, 0, 3, 3, 0, 0, 0, 0, 0, 0],
[0, 0, 0, 0, 3, 3, 0, 0, 0, 0, 0, 0], [0, 7, 7, 3, 3, 3, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 3, 3, 3, 0, 0, 0]]},
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"output": [[0, 0, 9, 0, 0, 0, 0, 0, 0, 0], [0, 0, 9, 0, 0, 0, 0, 0, 0, 0], [0, 0, 9, 0, 0, 0, 0, 9, 0, 0], [0, 0, 3,
0, 0, 0, 0, 6, 0, 0], [0, 0, 3, 0, 0, 0, 6, 0, 6, 0], [0, 3, 0, 3, 0, 0, 6, 0, 6, 0], [3, 0, 0, 0, 3, 0, 0, 0, 0, 0],
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0, 0], [0, 0, 0, 0, 0, 6, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 6, 6, 6, 0, 0, 0, 0, 0], [0, 0, 0, 6, 6, 0, 0, 6, 0, 0, 0,
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"output": [[0, 0, 0, 0, 0, 7, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 7, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 7, 0, 0, 0, 0,
0, 0], [0, 0, 0, 0, 0, 6, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 6, 6, 6, 0, 0, 0, 0, 0], [0, 0, 0, 6, 6, 0, 0, 6, 0, 0, 0,
0], [0, 0, 6, 0, 0, 0, 0, 0, 6, 0, 0, 0], [0, 0, 6, 0, 0, 0, 0, 0, 6, 0, 0, 0], [0, 0, 6, 0, 0, 0, 0, 0, 6, 0, 0, 0],
[0, 0, 6, 0, 0, 0, 0, 0, 6, 0, 0, 0], [0, 0, 6, 0, 0, 0, 0, 0, 6, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0]]}],
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7, 0, 9, 0, 0, 0, 0, 8, 0, 0, 0]],
"output": [[0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 7, 0, 2, 2, 2,
2, 2, 0, 0, 0], [0, 0, 0, 7, 0, 0, 0, 0, 0, 0, 2, 4, 4], [0, 0, 0, 7, 0, 2, 2, 2, 2, 2, 0, 0, 0], [0, 0, 0, 7, 0, 0,
0, 0, 0, 0, 0, 0, 0], [0, 0, 0, 9, 0, 0, 0, 0, 0, 0, 0, 0, 0], [0, 0, 9, 9, 0, 0, 0, 0, 0, 8, 0, 0, 0], [0, 9, 0, 0,
9, 0, 0, 0, 0, 0, 0, 8, 0, 0], [0, 9, 0, 0, 9, 0, 0, 2, 0, 0, 0, 8, 2], [0, 9, 0, 0, 9, 0, 0, 0, 0, 0, 8, 0, 0], [0, 9,
0, 0, 9, 0, 0, 0, 0, 8, 0, 0, 0]]
}]
}
```




ARC-AGI-1: static puzzles, now near saturation



Timeline

2019
Benchmark launched

Dec 2024
o3-preview 75.7%, or 87.5% at about 172x compute [12]

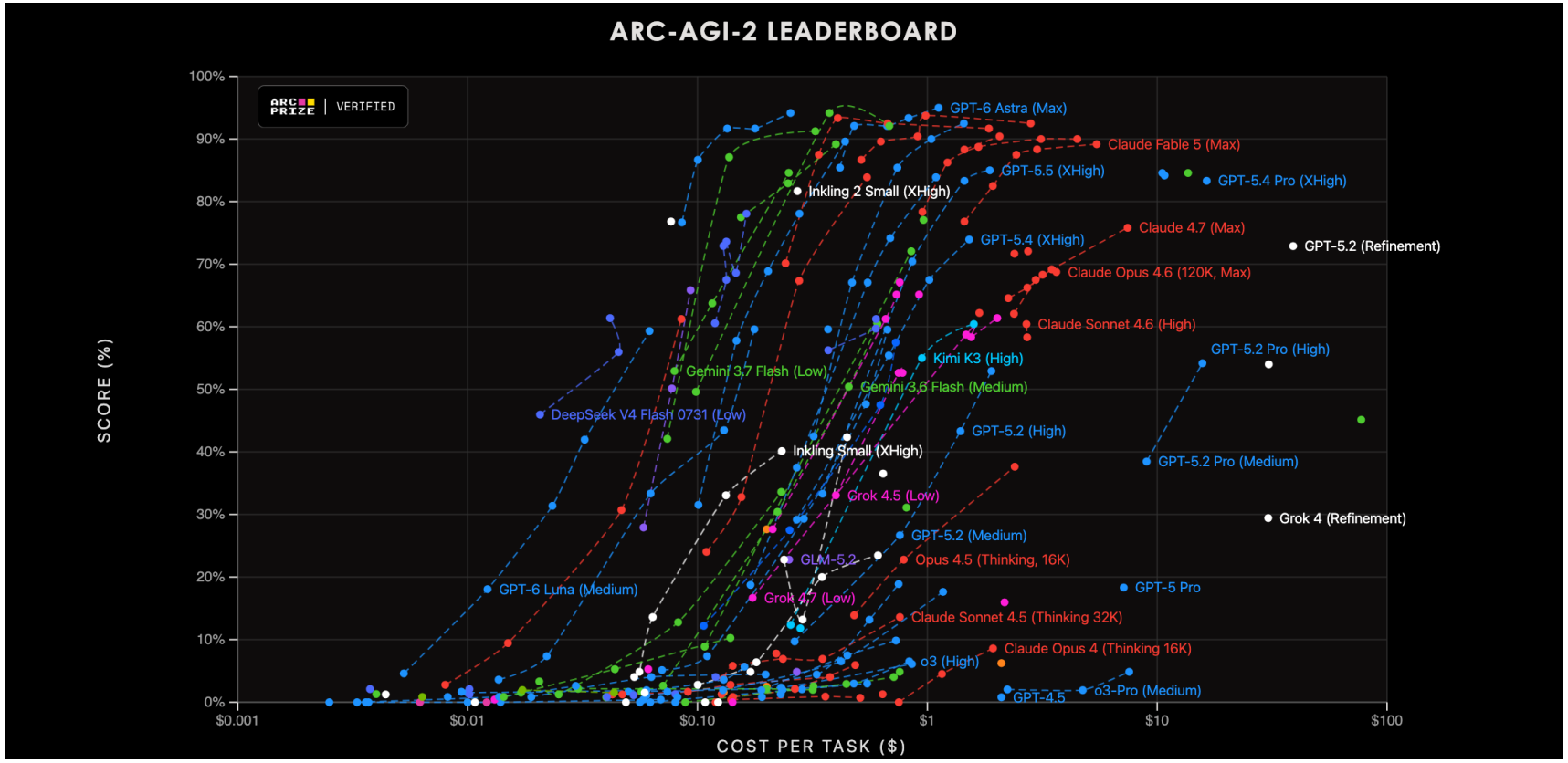
Apr 2025
o3 (medium) 57% at $1.5 per task [13]

Oct–Nov 2026
98.5% at $0.16–0.36 per task [5]

Read it as: a high score is only half the story. o3-preview was trained on most of the public training set [12], so cost and the evaluation split matter.

Sources: [5] ARC Prize posts on X; [12] o3 report summary; [13] ARC Prize analysis via press. Argon = Gemini 4 Argon. Check arcprize.org/leaderboard.

ARC-AGI-2: harder, with cost as a metric



Timeline

Mar 2025

Launch: pure LLMs about 0%; o3 (low) about 4% at about $200 per task [10, 11]

Nov 2025

Best Kaggle entry 24% at $0.20 per task [2]

Feb 2026

Gemini 3.1 Pro 77.1% (reported) [15]

Oct–Nov 2026

91–94% verified [5]

ARC-AGI-3: no instructions, no rules, no goal

Interactive, turn-based games handcrafted by game designers. The agent gets no instructions, no rules and no stated goal [1, 3].

Explore

Infer the goal

Model the dynamics

Plan and act

Scored on action efficiency: turns taken to complete each level on first sight, compared with a human baseline [1].

0.1–0.5%

best frontier models at launch, March 2026. Humans solved the environments [1].

Demo 1b: play one public game

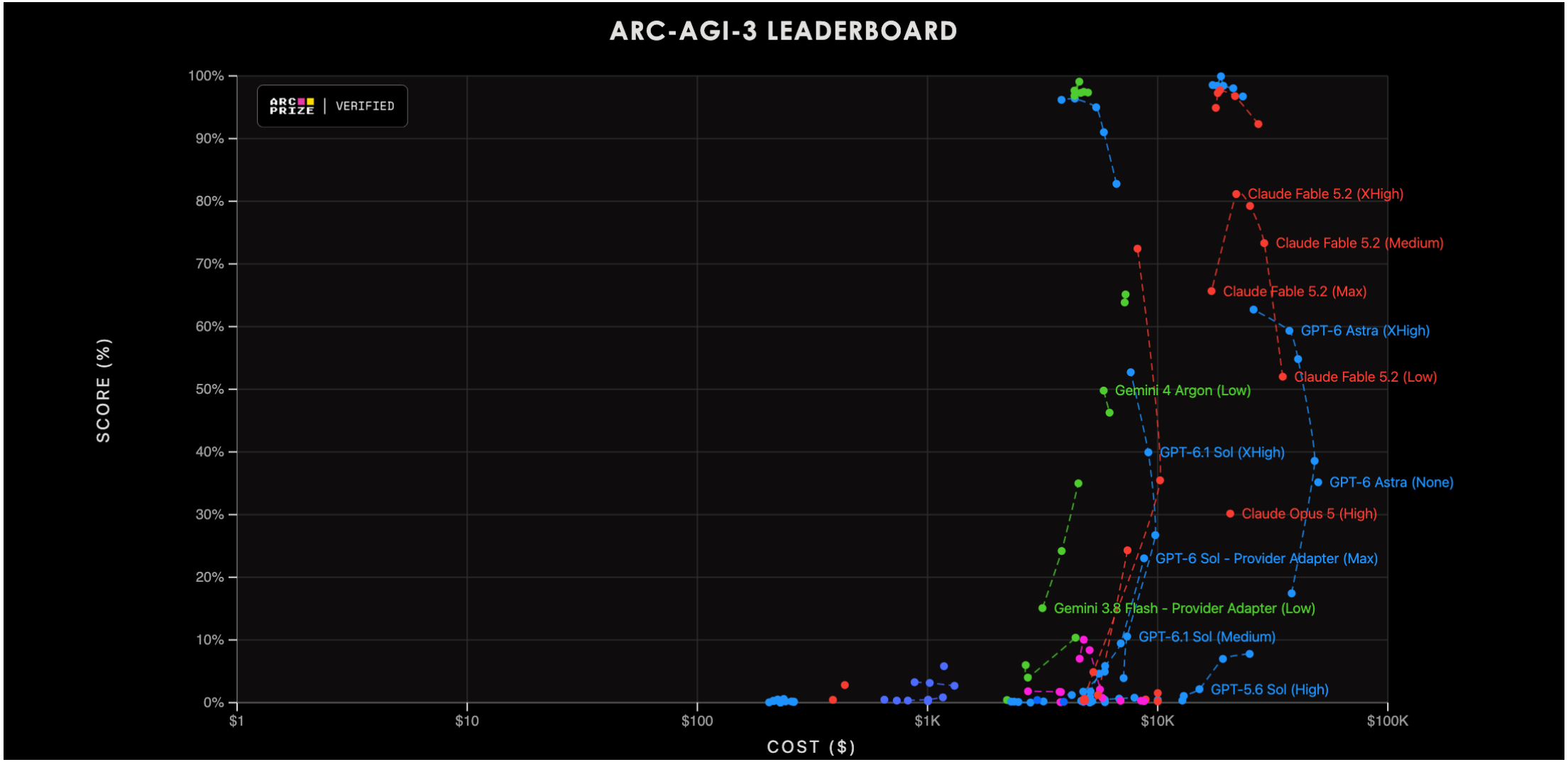
No instructions. Watch your first ten seconds: you press things, observe, and guess the goal. That is the capability being tested.

Day-one link

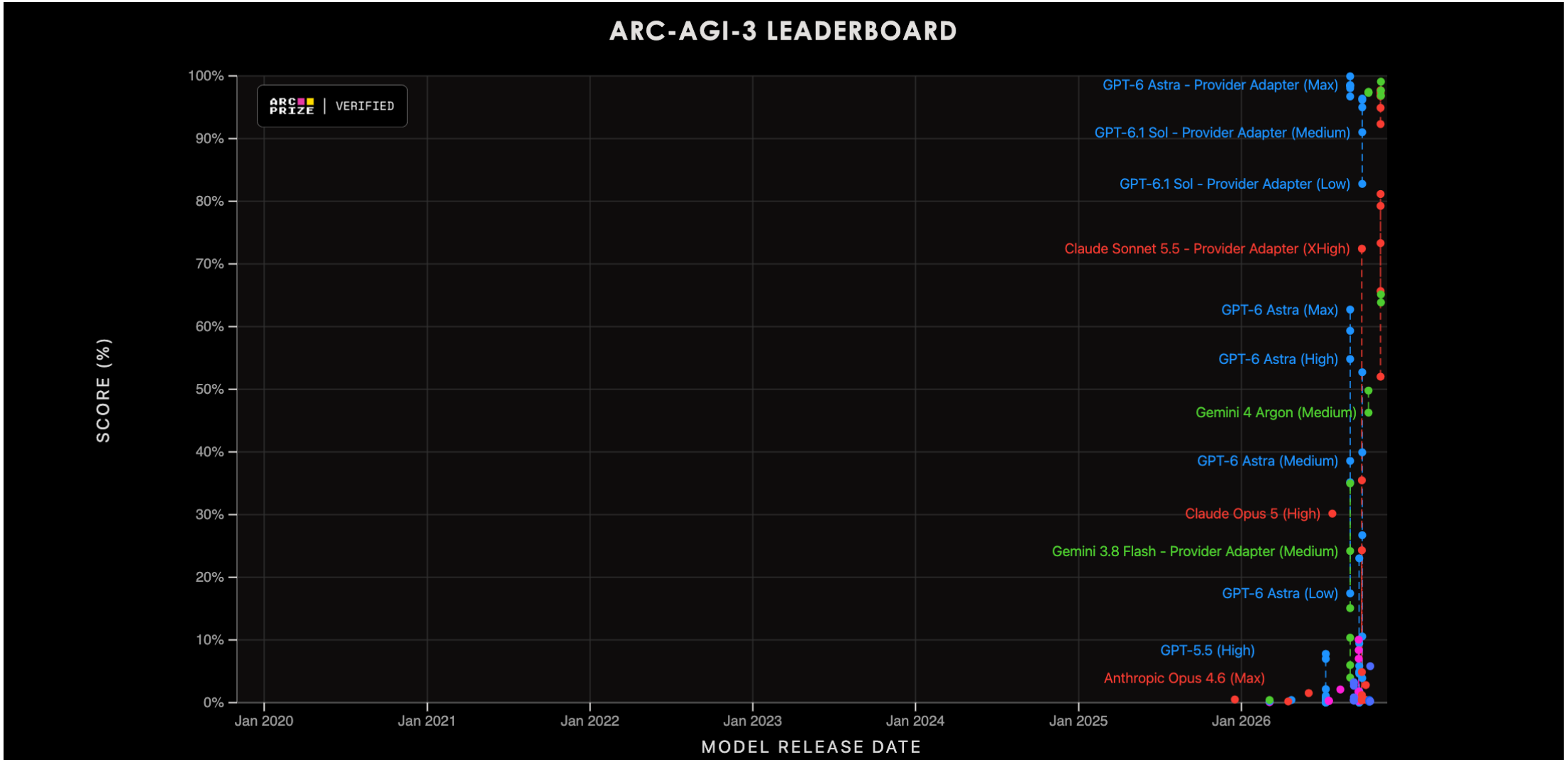
A new tool, no manual, an unknown goal. This is the closest benchmark to the use case.

Sources: [1] ARC-AGI-3 technical report (2026); [3] ARC Prize launch post.

ARC-AGI-3: no instructions, no rules, no goal



ARC-AGI-3: no instructions, no rules, no goal



Benchmark today: fast progress, with caveats

Verified (ARC Prize)	ARC-AGI-1	ARC-AGI-2	ARC-AGI-3 Standard	ARC-AGI-3 Prov. Adapter
Launch, Mar 2026 (best of four)	–	–	0.5%	–
GPT-6 Astra (Sep 2026)	–	–	62.7%	99.9%
Gemini 4 Argon (Oct 2026)	98.5%	91.3%	49.8%	97.5%
Claude Fable 5.2 (Nov 2026)	98.5%	93.8%	81.2%	97.7%

Harness matters

Standard: the model keeps its own notes. Provider Adapter: the provider keeps reasoning state and compacts history [4].

Cost is large

Full ARC-AGI-3 runs cost about $5K to $26K [4, 5].

Not like-for-like

Humans had no scratch pad or code tool; the authors do not claim AGI [4].

Next

ARC Prize 2026 results will be announced on 4 December [6].

Sources: [1] technical report; [4] ARC Prize blog (Astra); [5] ARC Prize X posts; [6] arcprize.org/competitions/2026. Check arcprize.org/leaderboard on the day.

Methodology: apply ARC-AGI to the use case

1
Define
the capability: infer a rule from a few examples

2
Build tasks
k labelled alerts, one new case; hold out whole typologies

3
Run
prompt, model, parse the answer

4
Score
exact match, 2 attempts, plus cost per task

5
Compare
vs rules, an ML model and an analyst

- ARC-AGI answers “can it adapt?”
- Run the official public tasks first.

- Your task set answers “does it work here?” Same protocol: few-shot examples, unseen test, automatic grading.
- Report both, with settings, split and cost.

- **If the agent task test cases encode as grids, the official harness runs them unchanged; otherwise reuse the protocol and scoring pattern.**

```
git clone https://github.com/arcprize/
  arc-agi-benchmarking
cd arc-agi-benchmarking && uv sync

uv run main.py --config <model-config> \
  --data_dir <tasks-dir> \
  --save_submission_dir submissions/run1
```




Key message and how to reach me

Don’t ask what your AI agent already knows.
Ask how fast it learns your world, and measure it.

Questions?

Thank you for attending this talk.

Learn and run

arcprize.org

ARC-AGI-3 paper

arc-agi-benchmarking

Chollet 2019

Contact

Govind Tatachari
CognitivSys LLC
Email: [your email]
Web: https://www.cognitivsys.com



ARC Prize

References

[1] ARC Prize Foundation, ARC-AGI-3 technical report (2026); Chollet, On the Measure of Intelligence (2019), arXiv:1911.01547

[2] Chollet et al., ARC Prize 2025 Technical Report, arXiv:2601.10904

[3] ARC Prize, Announcing ARC-AGI-3, 25 Mar 2026

[4] G. Kamradt, OpenAI's GPT-6 Astra on ARC-AGI-3, arcprize.org/blog/astra (3 Sep 2026)

[5] ARC Prize on X: Gemini 4 Argon (15 Oct 2026); Claude Fable 5.2 verified results (most recent post found)

[6] arcprize.org/competitions/2026

[7] Hilal, Gadsden, Yawney, Financial Fraud: A Review of Anomaly Detection Techniques and Recent Advances, Expert Systems with Applications 193 (2022)

[8] Yang, Shukur, Sahran, A Review of AI for Financial Fraud Detection, Applied Sciences 16(4):1931 (2026)

[9] Industry transaction-monitoring analyses: The Stack; Sphinx; SanctionScanner (citing McKinsey); figures vary, 85–95% false positives

[10] eWeek, ARC-AGI-2 benchmark, 27 Mar 2025

[11] The Decoder, ARC-AGI-2 initial results, Mar 2025

[12] ARC Prize o3 report (Dec 2024); summary at longtermwiki.com

[13] ARC Prize o3/o4-mini analysis (Apr 2025) via 36Kr press coverage

[14] GPT-5 ARC-AGI-2 result via press (Aug 2025)

[15] OfficeChai, ARC-AGI-3 released (Mar 2026), reporting Gemini 3.1 Pro 77.1% on ARC-AGI-2